Retail Banking Business Data Model: Conceptual and Logical Model, Business Glossary and Data Quality Rules.

Section 1. Purpose

This is a self-initiated project that builds a business data model for retail banking. It covers a conceptual model, a business glossary, a logical model and a set of data quality rules, and it ends with a note on how the work relates to data governance and BCBS 239. The aim is to show how a business concept moves from an agreed definition, to a model, to a testable data quality rule, with a named owner accountable for its meaning.

The project is a learning and demonstration exercise. It is not based on any real bank's internal models, and it does not represent the data model of any specific institution.

Section 2. Scope

In scope: retail banking customers, their accounts, and the transactions recorded on those accounts. The model covers eight business concepts: Party, Customer, Address, Product, Account, Account Holder, Transaction and Channel.

Out of scope:

- Guarantors and authorised signatories. A guarantor or signatory would also be a Party, but these roles are not modelled.
- Investors and investment products
- Mortgages and insurance
- Next of kin
- Counterparty (receiver) details
- Operational metadata about data movement, such as load times and ongoing load monitoring. A one-off check that the Berka row counts matched the source is reported in Step 4, but no load log or pipeline is modelled.

Section 3. Approach

The work is done in five stages. Each stage uses the output of the one before it.

1. **Conceptual model.** Business entities and their relationships, drawn in crow's foot notation. Each relationship is read in both directions to check that it makes business sense.
2. **Business glossary.** A definition, an assumed owner role and an example for each of the 8 entities and 17 key attributes. Definitions were tested against edge cases such as joint accounts, signatories and guarantors.
3. **Logical model.** Entities become tables and attributes become columns, with data types, primary keys and foreign keys.
4. **Data quality rules.** Rules R1 to R9, derived from the glossary and the logical model, tested on the Berka dataset. Two further findings, F1 and F2, came from profiling. Results are reported by data quality dimension. Synthetic rows are used only to check that the rules can detect errors.
5. **Write-up and governance.** The key modelling decisions, a note on how the work relates to BCBS 239, and a governance operating model. The operating model covers roles, decision rights, a metadata register, data lifecycle and classification, quality KPIs, compliance controls, data issue management, and a data quality and stewardship policy with its controls.

The stages were iterative. Changes to the model came from trying to write clean definitions and from review. For example, "holds" became "acts as" and later "is associated with" between Customer and Account Holder, and "can be a" became "can have" between Party and Customer.

The metadata produced in each stage is of a different kind. **The glossary is business metadata**: it says what each term means and who owns it. **The logical model is technical metadata**: it says how the data is stored, meaning table, column, type, key and whether the column is required. **The data quality results are operational metadata**, because they describe how the data behaves in use. The metadata register (Section 11.3) holds all three kinds in one place.

Section 4. Assumptions and scope boundaries

Each assumption is numbered A1 to A31 and tagged with the stage in which it was made. Choices with real alternatives and trade-offs are recorded separately as decisions (C, L and DQ numbers) in Section 8. Other sections cite assumptions by these numbers.

#	Stage	Assumption
A1	Conceptual model	The model covers retail accounts, customers and transactions only. Customer is an enterprise-wide concept, so a Customer may have no Account Holder record, for example when onboarded but not yet funded, or when the relationship comes from a product outside this scope.
A2	Conceptual model	A Party has at most one Customer relationship. A prospect becomes a Customer when the relationship is established and keeps the same Party identity. Customer status over time (prospect, active, dormant, closed) is not modelled.
A3	Conceptual model	A returning former Customer keeps the same Customer record.
A4	Conceptual model	Every Account has at least one Account Holder record. A key cannot enforce a minimum on the parent side, so this is tested as a data quality rule (A22).
A5	Conceptual model	Only the originating Channel is recorded for each Transaction, and each Transaction has exactly one Channel.
A6	Conceptual model	A transfer between accounts is recorded as one Transaction line per Account.
A7	Conceptual model	A Party has at least one Address, and each Address record belongs to one Party only. Two people sharing a home each have their own Address record.
A8	Conceptual model	Closed Accounts are kept in the data with the status “closed”. Only the current status is stored. The closure date and status history are not modelled.
A9	Glossary	Owner roles are assumed, described by function, and would be validated with the business. The owner shown for each term is the Data Owner (Section 11.1).
A10	Glossary	All definitions are working definitions. Legal wording varies by bank and country, so they would also be validated with the business.
A11	Glossary	Address means a physical address. Email addresses and phone numbers are contact details and are not modelled.
A12	Glossary	An Account Holder is a Customer named on the agreement for a specific Account, recorded as one Account Holder record per Customer and Account. Signatories and guarantors are excluded.
A13	Glossary	Each Account has one currency, and all its Transactions are recorded in that currency. The original foreign amount and exchange rate are not kept.

A14	Glossary	Transaction Amount is always positive. Direction is held in the Debit/Credit Indicator. Where a source system uses its own direction codes, a code is mapped to D or C only when the source documentation and the meaning of the transaction support it (A25).
A15	Glossary	Customer ID, Account Number (IBAN), postcodes, and country and currency codes are defined by the bank or an external standard, so their formats are taken as given. They are stored as text (L8).
A16	Logical model	The logical model is independent of any database product. Data types are generic (integer, text, decimal, date, timestamp), and indexes, partitioning and performance are not considered.
A17	Logical model	Every attribute in scope is needed to define its entity, so no column may be empty. This is a choice for this project's scope, and a real bank would allow optional columns (L6).
A18	Logical model	A Customer holds a given Account in one continuous period. Leaving and rejoining the same Account is not modelled, because no start or end date is stored (L1).
A19	Logical model	Allowed-value lists (Party Type, Address Type, Product Type, Channel Type, Account Status) are working lists, held as rules on the column, and would be agreed with the owners (L4).
A20	Logical model	Transaction Amount is stored with two decimal places. Currencies with a different number of decimal places are out of scope.
A21	Logical model	Transaction Timestamp is stored as one date and time, with no time zone. Time zone handling is not modelled.
A22	Logical model	Rules that a key cannot enforce are treated as data quality rules, not structure. Examples are every Party having at least one Address (A7) and every Account having at least one Account Holder record (A4). They are carried into Step 4.
A23	Data quality	Berka fields are mapped to this model, and the mapping is stated. Berka has dates but no times, no IBAN, no currency column, no names or addresses, and records transaction direction in its own way. Rules on timestamps are therefore tested at date level only.
A24	Data quality	A Customer with no Account Holder record is reported for review. It is not treated as an error. In Berka, all 869 such clients hold only the disponent role, and authorised signatories are out of scope (C3).
A25	Data quality	Berka type maps to the Debit/Credit Indicator: PRIJEM to C and VYDAJ to D. The value VYBER is not a documented value for type. It is mapped to D only as a target-model decision (DQ2), and the mapping does not make the source value valid.
A26	Data quality	Berka operation supplies the Channel Type values: card withdrawal, cash deposit, cash withdrawal, transfer in and transfer out. A sixth value, "internal posting", is added in DQ3.
A27	Data quality	Berka account_id stands in for the IBAN, which Berka does not have (DQ4).

A28	Data quality	Only OWNER rows in Berka disp map to Account Holder. DISPONENT rows are excluded.
A29	Data quality	Berka dates are stored as YYMMDD text. Raw tables keep them as text, and two-digit years are converted with the %y rule, which reads 69 to 99 as 19xx and 00 to 68 as 20xx. In Berka, account opening dates run from 1 January 1993 to 29 December 1997, and transaction dates run from 1 January 1993 to 31 December 1998 (Script 08, query 8.4). A date such as 300101 is therefore read as 2030 and counts as a date in the future. No Berka date has a two-digit year from 00 to 68. The 2030 reading applies only to the synthetic test row in Script 08.
A30	Data quality	Synthetic rows are used only to test that rules can detect errors. They are added to labelled copies of the tables and are never counted in the results.
A31	Data quality	Two transactions are treated as a potential duplicate when they match in every column except `trans_id`, including the balance after the transaction. Identical content does not prove a double posting, and it does not prove the records are legitimate. Each potential duplicate is investigated and the outcome recorded.

Section 5: Step 1: Conceptual model

Purpose. The conceptual model identifies the business concepts in scope and how they relate, without technical detail. It gives everyone a shared picture before any definitions or tables are written.

Method. Eight concepts were identified from the scope: Party, Customer, Address, Product, Account, Account Holder, Transaction and Channel. Each relationship was named with a verb and read in both directions ("Each X ... Y" and "Each Y ... X"). Where a sentence read awkwardly, the verb or the model was changed. Two relationships were reworded after review. Party to Customer changed from "can be a" to "can have", so that Customer reads as the relationship attached to a Party and not as a different kind of person or organisation. Customer to Account Holder changed from "holds" to "acts as" and then to "is associated with", because an Account Holder is a record of a role and a Customer does not literally act as a collection of records.

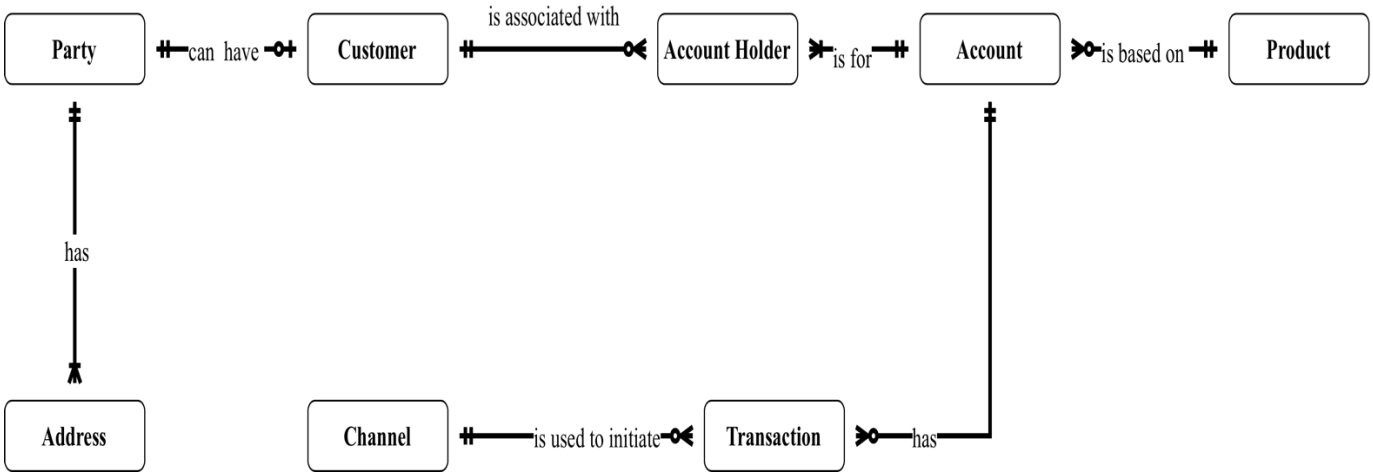


Figure 1: Conceptual model of retail banking, crow's foot notation.

How to read the diagram. Each end of a line has two marks. The mark nearest the box gives the maximum (bar = one, fork = many). The mark behind it gives the minimum (circle = zero, bar = at least one).

Conceptual Model Relationships.

Relationship	**Reading forward**	**Reading backward**
Party can have a Customer relationship	Each Party can have zero or one Customer relationship with the bank.	Each Customer is the established relationship of exactly one Party.
Party has Address	Each Party has one or more Addresses.	Each Address belongs to exactly one Party.
Customer is associated with Account Holder	Each Customer may be associated with zero or more Account Holder records.	Each Account Holder record relates to exactly one Customer.
Account Holder is for Account	Each Account Holder record relates to exactly one Account.	Each Account has one or more Account Holder records.
Account is based on Product	Each Account is based on exactly one Product.	Each Product is the basis for zero or more Accounts.
Account has Transaction	Each Account has zero or more Transactions.	Each Transaction belongs to exactly one Account.
Channel is used to initiate Transaction	Each Channel is used to initiate zero or more Transactions.	Each Transaction is initiated through exactly one Channel.

Identity, relationship and role. Party, Customer and Account Holder describe three different things.

- Party is identity: who the bank knows about.
- Customer is relationship status: who has an established relationship with the bank.
- Account Holder is a role: what a Customer is in relation to one specific Account.

What the model shows.

- **Party is broader than Customer.** A Party is any person or organisation the bank holds information about, including prospects, guarantors and payees. Their roles are out of scope, but they would still be Parties. Customer is not a separate kind of person or organisation. It is the established customer relationship that a Party has with the bank. A Party exists first and becomes a Customer when that relationship is established, without becoming a different Party. Only some Parties have a Customer relationship (A2).
- **Account Holder is a link entity.** Customer and Account have a many-to-many relationship. One Customer can be associated with many Accounts, and one Account can have many Customers, as in a joint account. Each Account Holder record links exactly one Customer to exactly one Account. This is a business statement. In the logical model, the Account Holder table resolves the many-to-many relationship.
- **Every Account has at least one Account Holder record (A4).** A foreign key and a composite primary key cannot enforce a minimum on the parent side. This is therefore tested as a data quality rule in Step 4, not as a structural constraint.
- **Product is the type and Account is the instance.** Many Accounts can be based on the same Product.
- **Transaction is the event and Channel is the route.** Each Transaction has exactly one originating Channel.

Assumptions and limits for this step. A1 to A8 in Section 4 apply to this step. The model records whether a Party has a Customer relationship, not how that status changes over time. It has no lifecycle stages and no relationship start or end dates. A more advanced model would add a customer status or relationship history entity, which is outside the scope of this learning model. These limits are repeated in Section 12.

Feeds next. The entities and relationships are defined in the business glossary (Step 2). The reasoning behind the main design choices is in Section 8, Key modelling decisions.

Section 6: Step 2: Business glossary

Purpose. The glossary fixes the meaning of each business term before any technical design. Each term has a definition, an owner and an example. The owner shown for each term is the Data Owner: the role accountable for the business definition, acceptable quality and remediation decisions for that term. Owners are **assumed roles**, described by function, and would be validated with the business. Data Stewards maintain the definitions day to day, as set out in Section 11.1. The glossary is **business metadata.**

Method. Definitions were drafted for the eight entities and seventeen key attributes, then tested against edge cases: joint accounts, signatories, guarantors, prospects, and customers with several accounts. This changed both the wording and the model. Account Holder was narrowed to exclude signatories and guarantors. Customer was redefined as the established relationship a Party has with the bank, not a separate kind of person or organisation. Account Holder was defined as a record of a role, so the relationship verbs were changed to match (see Step 1).

Entity glossary

Entity	Definition	Owner (assumed)	Example
Party	A person or organisation that the bank holds information about, whether or not it has a product with the bank.	Head of Customer Onboarding / KYC	A prospect who has requested a quote but holds no product.
Customer	The established relationship that a Party has with the bank. A Party exists before it becomes a Customer and keeps the same identity afterwards.	Head of Customer Onboarding / KYC	Anna de Vries, whose relationship with the bank was established when she opened a current account.
Address	A physical location associated with a Party, used by the bank to contact the Party or to meet legal and regulatory obligations.	Head of Customer Onboarding / KYC	Anna's home address, Keizersgracht 100, Amsterdam.
Product	A type of financial offering that the bank designs and sells to its customers.	Head of Product Management	"Student Current Account".
Account	A contractual arrangement between the bank and one or more Account Holders, under which the bank keeps a balance and records Transactions.	Head of Account Services	Anna's student current account with its own IBAN.

Account Holder	The role a Customer plays on a specific Account, recorded as one Account Holder record linking that Customer to that Account. The Account Holder is entitled to the Account's funds and responsible for its obligations. Signatories and guarantors are not Account Holders.	Head of Account Services	Anna and Ben on one joint account are two Account Holder records.
Transaction	A recorded movement of money into or out of an Account at a point in time.	Head of Payments	A €25 card payment on 3 March.
Channel	The means through which a Transaction is initiated.	Head of Digital / Distribution	Mobile app, ATM, branch.

Attribute glossary. Each attribute has the same assumed owner (the Data Owner) as its entity.

Entity	Attribute	Definition	Example	Rule
Party	Party Type	Whether the Party is a person or an organisation.	Person	"Person" or "Organisation"; not empty.
Party	Party Name	The full legal name of the Party, as shown on its identity or registration document.	Anna de Vries	Not empty.
Customer	Customer ID	A unique identifier assigned by the bank to a Customer.	C000123	Unique, not empty, never reused; stored as text.
Address	Address Type	The purpose of an Address, such as residential, postal or registered office.	Residential	One of the agreed values; not empty.
Address	Street	The street name and house number of the Address.	Voorbeeldstraat 12	Not empty.
Address	Postcode	The code assigned to the Address by the national postal service.	2511 AB	Not empty; stored as text; follows the postcode format of the Address's Country.
Address	City	The town or city of the Address.	Den Haag	Not empty.
Address	Country	The country of the Address (ISO 3166-1 alpha-2).	NL	Two capital letters from the ISO list; not empty.
Product	Product Type	The category of a Product, such as savings or current account.	Savings account	One of the agreed values; exactly one per Product.
Account	Account Number (IBAN)	The unique identifier of an Account.	NL91ABNA0417164300	Unique, not empty; stored as text without spaces.

Account	Account Opening Date	The date on which the Account was opened.	2024-03-15	Valid date; not in the future.
Account	Account Currency	The currency of the Account, in which all its Transactions are recorded (ISO 4217).	EUR	Three capital letters from the ISO list; one per Account.
Account	Account Status	Whether the Account is currently open or closed.	open	"open" or "closed"; not empty; one per Account.
Transaction	Transaction Amount	The value of money moved, always positive, in the currency of the Account.	25.00	Greater than zero.
Transaction	Debit/Credit Indicator	Whether the Transaction moves money out of the Account (debit) or into it (credit). This is the target model's indicator. Where a source system uses its own codes, a code is mapped to D or C only when the source documentation and the meaning of the transaction support it. A mapping does not make an undocumented source value valid.	D	"D" or "C" only; not empty.
Transaction	Transaction Timestamp	The date and time at which the Transaction occurred.	2026-03-03 14:22	Valid; not in the future; its date is not before the Account Opening Date of its Account.
Channel	Channel Type	The kind of means through which a Transaction was initiated.	Mobile app	One of the agreed values; exactly one per Transaction.

Key distinctions.

- **Party vs Customer:** Party is identity, meaning anyone the bank holds information about, including prospects, guarantors and payees. Customer is the established relationship a Party has with the bank. It is not a separate kind of person or organisation.
- **Customer vs Account Holder:** Customer is a Party's established relationship with the bank. An Account Holder record is that Customer's role on one specific Account. One Customer can be associated with several Account Holder records, and one Account has one or more of them.

- **Product vs Account:** Product is the type. Account is one instance of it, agreed with one or more Account Holders.
- **Transaction vs Channel:** Transaction is what happened. Channel is where it started.

Feeds next. The entities become tables and the attributes become columns in the logical model (Step 3). The rules become the basis for the data quality rules in Step 4. The rule that a Transaction is not dated before its Account's opening date is defined once, under Transaction Timestamp, and tested once in Step 4 (R3).

Section 7: Step 3: Logical model

Purpose. To turn the conceptual model into a structure that could be stored without ambiguity. The conceptual model says what the business talks about. The logical model says how each fact and each relationship is held as data, while staying independent of any database product.

Method. Each entity became a table and each attribute a typed column. Each relationship became a foreign key on its "many" side, and the many-to-many between Customer and Account became the link table Account_Holder. Where an entity had no usable business identifier, a technical key was added. Each design choice with real alternatives is recorded as a decision (L1 to L9, Section 8), and each point taken as given is recorded as an assumption (A16 to A22, Section 4). The logical model uses underscores in names (Account_Opening_Date), and the glossary uses business names with spaces (Account Opening Date). The metadata register (Section 11.3) holds both, and the term is the link.

Key terms

Term	Meaning
Table	The storage for one entity. Each entity becomes one table.
Column	One kind of fact held in a table. Every attribute becomes a column.
Row	One instance of what the table holds, for example one Account or one Transaction.
Primary key (PK)	The column or columns that uniquely identify each row in a table.
Foreign key (FK)	A column that holds the primary key value of a row in another table, which is how a relationship is stored.
Surrogate key	A system-generated identifier that identifies a table's own rows and has no business meaning, such as Account_ID.
Natural key	An identifier defined by the business or an external standard, such as Customer_ID or the IBAN. Contrast with a surrogate key, which the model generates.

Technical key	A key column added by the modeller rather than defined by the business. It covers surrogate keys and foreign keys, including a foreign key that copies a natural key.
Constraint	A rule built into the structure that stops bad data being stored (primary key, foreign key, unique, required).

Logical model relationships

Relationship	**FK column**	**Sits in table**	**Points to**
Party can have a Customer relationship	Party_ID (also unique)	Customer	Party
Party has Address	Party_ID	Address	Party
Customer is associated with Account Holder	Customer_ID	Account_Holder	Customer
Account Holder is for Account	Account_ID	Account_Holder	Account
Account is based on Product	Product_ID	Account	Product
Account has Transaction	Account_ID	Transaction	Account
Channel is used to initiate Transaction	Channel_ID	Transaction	Channel

The relationships and their cardinalities are unchanged from the conceptual model. The Step 1 table is the source for the business reading, and this one is its technical counterpart.

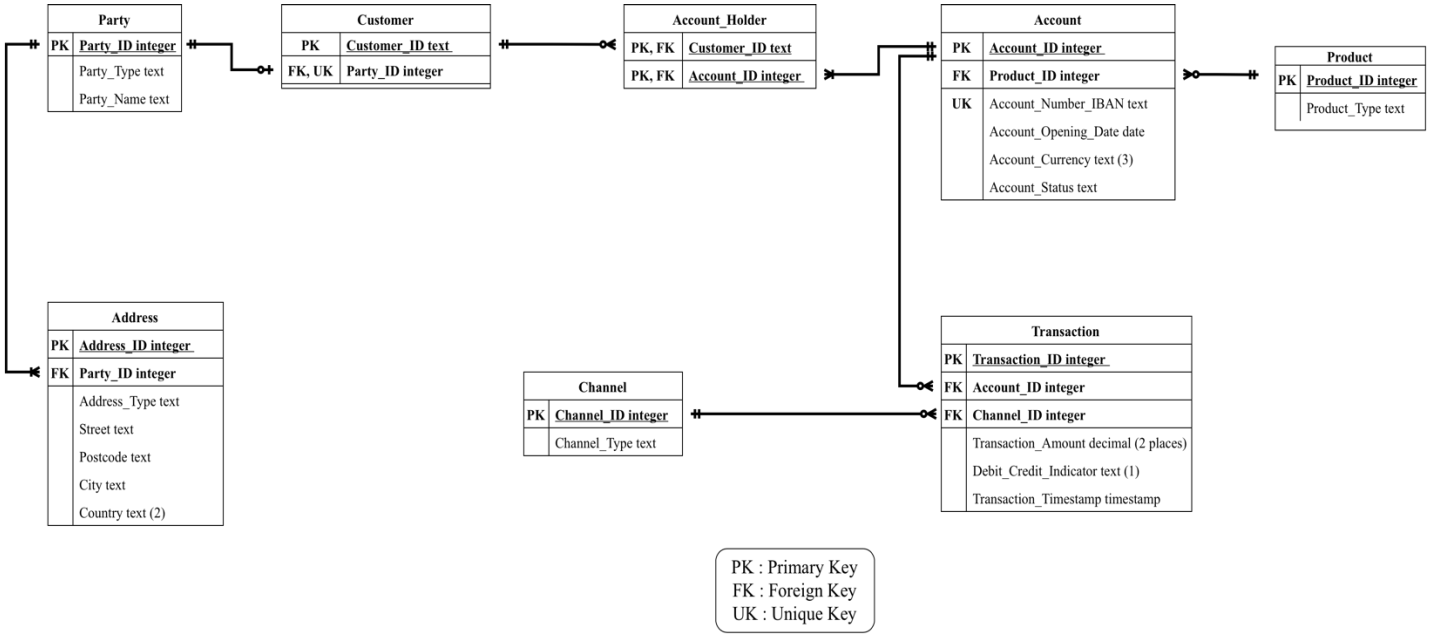


Figure 2: Logical model of retail banking, crow's foot notation.

The marks show the business cardinalities from Step 1. The structure does not enforce the minimums "every Account has at least one Account_Holder row" (A4) and "every Party has at least one Address" (A7). They are tested as data quality rules (R9, and the Address rule that Berka cannot run).

Technical metadata (logical model table).

Each column maps to its business term in the glossary, which is how the business and technical metadata are linked.

Table	Column	Type	Key	Required?
Party	Party_ID	integer	PK	Yes
	Party_Type	text		Yes
	Party_Name	text		Yes
Customer	Customer_ID	text	PK	Yes
	Party_ID	integer	FK, UK	Yes
Address	Address_ID	integer	PK	Yes
	Party_ID	integer	FK	Yes
	Address_Type	text		Yes
	Street	text		Yes
	Postcode	text		Yes
	City	text		Yes
	Country	text (2)		Yes

Product	Product_ID	integer	PK	Yes
	Product_Type	text		Yes
Account	Account_ID	integer	PK	Yes
	Account_Number_IBAN	text	UK	Yes
	Product_ID	integer	FK	Yes
	Account_Opening_Date	date		Yes
	Account_Currency	text (3)		Yes
	Account_Status	text		Yes
Account_Holder	Customer_ID	text	PK, FK	Yes
	Account_ID	integer	PK, FK	Yes
Channel	Channel_ID	integer	PK	Yes
	Channel_Type	text		Yes
Transaction	Transaction_ID	integer	PK	Yes
	Account_ID	integer	FK	Yes
	Channel_ID	integer	FK	Yes
	Transaction_Amount	decimal (2 places)		Yes
	Debit_Credit_Indicator	text (1)		Yes
	Transaction_Timestamp	timestamp		Yes

- Allowed-value rules and other business rules are held in the glossary (Rule column).
- Technical key columns have no glossary entry because they carry no business meaning.

Constraints

Constraint	**Meaning**	**Example in the model**
Primary key (PK)	Unique and never empty	Account_ID in Account
Foreign key (FK)	Must point to an existing row	Transaction.Account_ID must match a real Account
Unique (UK)	No two rows may share the value	Account_Number_IBAN in Account
Required	Never empty	Every column (decision L6)

What the model shows.

- Eight tables and seven foreign key relationships, one for each line of the conceptual model.
- Thirty columns in total: the 17 business attributes from the glossary and 13 technical key columns (6 surrogate keys and 7 foreign keys).
- Five tables hold a foreign key (Customer, Address, Account, Account_Holder, Transaction). Party, Product and Channel hold none, because they only ever sit on the “one” side.

- Uniqueness built into the structure: one Customer relationship per Party (unique Party_ID in Customer), one row per Customer and Account pair (composite key in Account_Holder), and one row per IBAN (unique Account_Number_IBAN).
- Every column is required, and every key and type decision is traceable to a logged decision.
- Some rules cannot be enforced by the structure. A key can stop a bad link, but it cannot force a parent row to have a child row. Every Account must have at least one Account_Holder row (A4), and every Party must have at least one Address (A7). These are tested as data quality rules.
- A new product type is added by approving a new value in the Product Type list (decision L4) and adding a row to Product. No new table or column is needed.

Feeds next. The structure supports the Step 4 rules. The rules that a key cannot enforce become tested data quality rules: every Account has at least one Account_Holder row (R9), and no Transaction is dated before its Account's opening date (R3). The rule that every Party has an Address is defined but cannot be run on Berka, which has no address data. The column names and types also give the Berka mapping its target.

Section 8: Key modelling decisions

#	Decision	Options considered	Choice and reason
C1	Can a Customer exist without an Account?	One or many vs zero or many	Zero or many, because Customer is enterprise-wide (A1).
C2	How to link Customer and Account	Direct line vs Account Holder link entity	Link entity, to resolve joint accounts (many-to-many). The relationship is worded "Each Customer may be associated with zero or more Account Holder records", because an Account Holder is a record of a role.
C3	Who counts as an Account Holder	Include signatories and guarantors vs exclude	Exclude. They are out of scope and noted as future roles (A12).
C4	Party to Customer cardinality	Zero or one vs zero or many	Zero or one, to prevent duplicate customers of the same Party. Customer is the established relationship of an existing Party, so a prospect becomes a Customer without becoming a new Party. Trade-off: lifecycle status and history are not modelled (A2). A more advanced model would add them.
C5	Where currency lives	On Transaction vs on Account	On Account. Trade-off: the foreign amount and exchange rate are lost (A13).
C6	How direction of money is stored	Negative amounts vs positive amount plus Debit/Credit Indicator	Positive plus indicator, to avoid sign errors and allow a clear rule (A14).
C7	Which channel is recorded	Originating only vs every channel touched	Originating only (A5).
C8	How transfers are recorded	One transaction vs one line per account	One line per account (A6).
C9	Where names and addresses live	On Customer and Account Holder vs on Party only	Party only, so no fact is stored twice.
C10	How to enforce at least one Account Holder per Account	Structural constraint vs data quality rule	Data quality rule (R9). A foreign key and a composite key cannot enforce a minimum on the parent side (A4, A22).
L1	How to identify an Account Holder row	Composite key (Customer_ID + Account_ID) vs own ID plus a unique constraint on the pair	Composite key, so the structure itself stops the same pair appearing twice. Nothing else points at Account_Holder, so a two-column key costs nothing. Trade-off: no leave-and-rejoin history (A18).

L2	Which column is the primary key of Account	Account Number (IBAN) vs a system-generated Account_ID	Account_ID, with the IBAN kept as a unique, required column. The IBAN would otherwise be copied into every Account_Holder and Transaction row, and a change would have to be made everywhere. Berka also has an account ID and no IBAN. Trade-off: one extra technical column.
L3	Whether Customer needs a surrogate key	Add a surrogate key vs use Customer_ID	Use Customer_ID. It is already unique, bank-assigned, never reused and recognised by the business, so a second ID would add nothing. Trade-off: text keys are slightly bulkier than integers.
L4	How to store attributes with a fixed list of values	Separate reference tables vs text columns with an allowed-value rule	Text columns with a rule. The lists are short, and Channel and Product are already near-reference tables. Trade-off: adding a value means changing the rule. A real bank would likely use governed reference tables. A new product type is added by approving a new value in the Product Type list and adding a row to Product. No new table or column is needed.
L5	How much detail Address holds	Type only vs Type plus Street, Postcode, City and Country	Add the four, because the Address definition requires a real location and Country allows an ISO rule. Trade-off: the glossary grows by four attributes.
L6	Whether columns may be empty	Allow optional columns vs all required	All required (A17). Every attribute in scope is needed to define its entity, and it keeps the Step 4 rules simple. Trade-off: stricter than most real banks.
L7	How closed Accounts are shown	Reword the assumption only vs add an Account Status column	Add Account Status (open or closed), so closed Accounts stay in the data and can be told apart (A8). Trade-off: current status only, with no closure date or history. The glossary now has 17 attributes.
L8	Which data type identifiers get	All integers vs all text vs split by who defines the format	Model-generated identifiers are integers, because the model controls their format. Business or external identifiers and codes (Customer ID, IBAN, postcode, country and currency codes) are text, because their format may include letters, spaces or leading zeros (A15). Trade-off: if an external format changes, the model must be reviewed.
L9	Whether to model Transaction Type	Add it as an 18th attribute vs leave it out	Leave it out. Debit/Credit Indicator and Channel already hold direction and means, and no quality

			rule uses it. Trade-off: a known simplification, and Berka's transaction type fields are not mapped as a separate attribute.
DQ1	How to treat transactions with a zero amount	Keep "greater than zero" vs change to "zero or more" vs exempt interest postings	Keep "greater than zero" and report the 14 as failures, because a zero line moves no money, and a looser rule would also accept missing amounts filled in as zero.
DQ2	How to map the 16,666 rows with VYBER in type	Map to D vs leave unmapped vs exclude	Map to D as a target-model decision, and keep the rows reported as a source finding (F1). The documentation lists VYBER under operation ("withdrawal in cash") but not under type. The mapping is supported by three facts: all 16,666 rows have operation VYBER, the balance falls by exactly the amount on all 9,597 rows whose position in the sequence is certain, and direction is needed to read a positive amount (C6). Trade-off: the mapping is an inference, not a documented value. The balance evidence covers 57.6% of the rows, and it should be confirmed with the source owner. R7 depends on this mapping.
DQ3	Channel for credits with an empty operation	Add "internal posting" vs leave empty vs make Channel optional	Add "internal posting" to the Channel Type list, because all 183,114 rows are interest credits created by the bank, and a named value keeps every transaction explainable, which an empty Channel does not.
DQ4	What stands in for the IBAN, which Berka lacks	Use account_id vs generate IBANs vs drop the rule	Use account_id, because it tests uniqueness and completeness without inventing data.
DQ5	How to describe identical transactions	Call them duplicates vs call them potential duplicates	Potential duplicates (A31). Identical content does not show that a record was posted twice, and it does not show that both records are legitimate. Each pair is investigated and the outcome recorded.
DQ6	Which dimension the "not in the future" check belongs to	Timeliness vs Validity	Validity (R8). The check tests whether a date is plausible. Timeliness is the delay between an event and its recording, which Berka cannot show because it has no load or posting times.
DQ7	Tolerance in the balance check	Exact comparison vs an agreed tolerance	Exact comparison (R7), because no tolerance has been agreed with an owner. The size of each

			difference is reported separately, so an owner can later decide whether a tolerance is acceptable.

Section 9: Step 4: Data quality rules and results

Purpose. To show that the glossary rules can be run on real data, and to report what they find by data quality dimension. The rules come from the Step 2 glossary and the Step 3 hand-offs, not from the dataset. The results show what each check establishes and what it does not.

ID	Table	Column tested (Berka field)	Rule	Dimension	Checked	Failed / flagged	%
R1a	Account	Account_ID (account_id)	Is unique	Uniqueness	4,500	0	0
R1b	Account	Account_ID (account_id)	Is not empty	Completeness	4,500	0	0
R2	Transaction	Transaction_Amount (amount)	Greater than zero	Validity	1,056,320	14	0.0013
R3	Transaction, Account	Transaction_Timestamp (date), Account_Opening_Date (date)	Not before opening date	Consistency	1,056,320	0	0
R4	Transaction	Debit_Credit_Indicator (type)	Every source value maps to D or C	Validity (target mapping)	1,056,320	0	0
R5	Customer, Account_Holder	Customer_ID (client_id, disp.type)	Has at least one OWNER (review population)	Completeness	5,369	869 flagged	16.19
R6	Transaction	All content columns	No two transactions are identical in every column except trans_id. Matches are reported as potential duplicates	Uniqueness	1,056,320	1 pair	0.0001
R7	Transaction	Transaction_Amount (amount), Debit_Credit_Indicator (type), balance	Balance equals previous balance plus signed amount	Consistency	406,941	5,296	1.30
R8	Transaction, Account	Transaction_Timestamp (date),	Not in the future	Validity	1,060,820	0	0

		Account_Opening_Date (date)					
R9	Account, Account_Holder	Account_ID (account_id, disp.type)	Has at least one OWNER	Completeness	4,500	0	0
F1	Transaction	Debit_Credit_Indicator (type)	`type` holds only the documented values, PRIJEM and VYDAJ	Validity (source)	1,056,320	16,666 (all VYBER)	1.58
F2	Transaction	Channel (operation)	Is not empty	Completeness	1,056,320	183,114	17.34

Source data. This used the public Berka dataset, which has account, client, disposition and transaction tables. All four files were loaded into DuckDB as untouched text copies (Script 01). This was done so that nothing was converted or guessed on the way in, and so that invalid values stayed visible. A one-off check confirmed that the row counts matched the source: 4,500 accounts, 5,369 clients, 5,369 dispositions and 1,056,320 transactions.

Method.

- Each table was profiled before any rule was written (Scripts 02 and 03). For example, profiling showed that trans.type has a third value, VYBER, that the documentation does not list under type. It appears only under operation.
- Berka fields were then mapped to the model (A23 to A31).
- Each rule was run as one query and the results were stored in one table (Script 04, which includes R9). R7 is in Script 07 and R8 is in Script 08.
- Rules that found nothing were tested again on labelled synthetic rows in copies of the tables (Scripts 05 and 08).

Results. R1 to R9 are rules written from the glossary before the data was examined. F1 and F2 are findings that profiling turned up, and they then measured them.

How to read the counts.

- *Checked* is the number of records the rule was run on.
- *Failed* is the number that broke the rule. For R5 the figure is *flagged for review*, because those records are not errors.
- For R6, the figure counts pairs of matching rows, not single rows, so one pair is two rows.
- The % column is Failed divided by Checked.

Reading the results.

- **R1a and R1b (Account ID is unique and not empty).** All 4,500 Account IDs are unique and none is empty. Berka has no IBAN, so `account_id` stands in for it (DQ4).
- **R2 (amount is greater than zero).** 14 of 1,056,320 transactions have an amount of exactly 0.00. According to their `k_symbol`, all 14 are interest postings (10 sanction interest, 4 interest credited).
    - A zero-amount interest line can be normal in the source system.
    - It still breaks the rule, because a Transaction is defined as a recorded movement of money. The 14 are reported as failures, and the Data Owner decides what to do with them (DQ1).
    - *Separate observation, not part of R2:* ten of the 14 carry the operation "cash withdrawal" although their `k_symbol` marks them as interest. This was looked at only on these 14 rows, not across the whole table.
- **R3 (no transaction before its account opened).** None of the 1,056,320 transactions is dated before the opening date of its account. The same check showed that every transaction matches an existing account, so none is orphaned.
- **R4 and F1 ask two different questions.**
    - *R4: can every value in `type` be mapped to D or C?* Yes, 0 failed. This passes by design, because the mapping covers every value found in the data. It shows that the mapping is complete. It does not show that the source values are valid.
    - *F1: does `type` hold only the values the documentation allows?* No. The documentation allows PRIJEM and VYDAJ. 16,666 rows (1.58%) hold VYBER, which the documentation lists only under `operation`, as "withdrawal in cash".
    - *Why VYBER is mapped to D (DQ2).* All 16,666 rows have the operation VYBER. On the 9,597 rows whose position in the sequence is certain, the balance falls by exactly the amount in every case. This is a decision about our model. It does not make the source value documented, and the source owner should confirm it.
- **R5 (every client has at least one OWNER record).** 869 of 5,369 clients (16.19%) have none. All 869 hold only the disponent role. A disponent is a user authorised to operate an account without being its owner, which is comparable to an authorised signatory and is out of scope (A24, C3).
- **R6 (no two transactions with identical content).** The check compares every column except `trans_id`, because `trans_id` is different on every row and so can never reveal a duplicate.
    - *Result:* one pair, 3456218 and 3507085. Both are on account 9051, dated 28 February 1998, type PRIJEM, with an empty operation, amount 0.00, balance 1,367.10 and `k_symbol` UROK. The bank and account fields are empty on both. They differ only in `trans_id`.

    - *What it means:* the pair is called a potential duplicate. The data shows identical content, but it cannot show whether this is one posting recorded twice or two genuine postings. The amount is zero and the balance did not change, so no money is affected either way. Deciding which it is would take the source system or the owner (A31, DQ5).
    - Both rows are among the 14 R2 failures, so the two findings overlap.
- **R7 (the balance follows from the previous balance).** After each transaction, the balance should equal the previous balance plus the amount for a credit, or minus the amount for a debit. The comparison is exact, with no tolerance (DQ7).
    - *Which rows are tested.* Berka has no times, so the order of same-day transactions is unknown. The rule runs only on transactions that are alone on their date and follow one that was also alone on its date. The first row of each account is excluded. That gives 406,941 rows, which is 38.5% of 1,056,320.
    - *Result:* 5,296 rows (1.30%) have a balance that does not match. Each is off by 0.50 or less. This is a count of the failures, not a tolerance.
    - *Cause:* not established. Rounding in the source is one possible explanation, to be confirmed.
    - The rule treats VYBER as a debit, so it relies on the DQ2 mapping. The result is in the Script 07 output, not in the Script 04 table.
- **R8 (no date is in the future).** 1,060,820 dates were checked: 1,056,320 transaction dates plus 4,500 account opening dates. None is in the future. R8 is a validity check, because it tests whether a date is plausible. It says nothing about how quickly data was recorded, so timeliness was not measured (DQ6).
- **R9 (every account has at least one OWNER record).** None of the 4,500 accounts is without one. Every Berka account has exactly one OWNER, so there are no joint accounts in this data.
- **F2 (Channel is not empty).** 183,114 transactions (17.34%) have an empty `operation`, so they have no Channel. All of them are interest credits created by the bank. The direction is known (credit), and the channel is what is missing. DQ3 adds the Channel Type "internal posting" so that each has a channel, and F2 would then fall to zero. That is a decision in the model, and the source does not say it. F2 is unrelated to R2, which tests the amount.

Testing the rules

R1a, R1b, R3, R4, R8 and R9 found no failures, which could mean the data is clean or that the rule cannot detect anything. Seven labelled synthetic rows were added to copies of the tables (Script 05):

- Two account rows with ID 999001, to break R1a (duplicate ID).
- Two account rows with an empty ID and a blank ID, to break R1b.

- One valid account, 999002, opened on 1 January 1997, which the transactions below use. It is designed to cause no failure.
- Transaction 9000001, dated 1996, before that account opened, to break R3.
- Transaction 9000002, with type XYZ, to break R4.

The rules detected the six rule-breaking rows as 1, 2, 1 and 1 failures for R1a, R1b, R3 and R4.

Two more rules were tested separately:

- R8 was tested in Script 08 with an eighth row, transaction 9000003, dated 1 January 2030. The rule flagged it (1 failure).
- R9 was tested with a synthetic account that has no OWNER row. The rule flagged it (1 failure).

So the zeros for R1a, R1b, R3, R4, R8 and R9 on the Berka data are meaningful. R2, R5, R6, R7, F1 and F2 were not tested synthetically, because they already found failures in the real data. The raw tables were never changed.

Limitations.

- Berka has dates but no times, no IBAN and no currency. Timestamp rules run at date level, and the IBAN and currency rules could not run.
- Timeliness, meaning the delay between an event and its recording, was not measured, because Berka has no load or posting times.
- Party Name, Address, Product Type and Account Status have no matching Berka fields and were not tested. The rule that every Party has an Address cannot be run.
- Every Berka account has exactly one OWNER, so the data never exercises joint accounts, the many-to-many Account Holder link, or an “at least one” rule beyond one.
- R1a counts a missing ID as a duplicate as well. This has no effect here, since no IDs are missing.
- R4 passes by construction. It shows that the mapping is complete, not that the source values are valid.
- R7 covers 38.5% of transactions, because the order of same-day transactions cannot be determined without times. It uses the Berka balance field, which the model does not hold, and it relies on the VYBER mapping.

- The dataset is from 1999, so the results show how the rules behave. They say nothing about the data quality of any real bank.
- Owners, thresholds and sign-off for each rule belong to the governance model (Section 11).
- The mismatch between `operation` and `k_symbol` was examined only on the 14 R2 failures.
- Ten of the 17 glossary attributes could not be tested, because Berka has no matching field: Party Type, Party Name, Address Type, Street, Postcode, City, Country, Product Type, Account Currency and Account Status. Customer ID was profiled but not tested, and Account Number was tested using account_id as a stand-in.

Section 10: BCBS 239 mapping table

Purpose. BCBS 239 is the Basel Committee's set of principles for **risk data aggregation and risk reporting** (2013). Its aim is that a bank can bring together its risk data accurately and quickly, and trust what it reports. This note shows how the foundations built in this project (definitions, ownership, structure and quality rules) support those principles.

Mapping of BCBS 239 themes to project evidence

BCBS 239 theme	What this project shows	Type of evidence	Evidence
Governance	Each entity and attribute has an assumed owner role, with decision rights and sign-off for changes. Data issues are logged and decided by owners (11.7), under a draft policy with named controls (11.8).	Designed	Glossary; 11.1, 11.2, 11.7, 11.8
Data architecture	Conceptual and logical models with agreed definitions, tables, keys, data types and constraints in one structure that data can be aggregated from.	Designed	Steps 1 and 3
Accuracy and integrity	Constraints stop some bad data at storage. Step 4 rules cover the rest: 14 of 1,056,320 amounts not above zero, one potential duplicate pair, 5,296 of 406,941 balances that do not reconcile (all by 0.50 or less, cause not confirmed), and 16,666 source direction values that the documentation does not list under type. The zero-failure rules R1a, R1b, R3, R4, R8 and R9 were shown able to detect errors on synthetic rows.	Measured	Step 3; Step 4 (R2, R4, R6, R7, F1, Scripts 05 and 08)

Completeness	Account Number (stood in for by account_id) not empty: no failures. Every Account has an Account Holder: no failures, and the rule was shown able to detect a missing holder, although Berka has exactly one owner per account. 869 customers (16.19%) with no Account Holder form a review population, not an error count. 183,114 transactions (17.34%) had no channel in the source, fix designed to include "internal posting".	Measured	Step 4 (R1b, R5, R9, F2)
Consistency across the bank	One shared glossary, tested against joint accounts, signatories and guarantors. No transaction is dated before its account opened.	Designed (glossary); measured (R3)	Step 2; Step 4 (R3)
Traceability	Each column maps to a glossary term and its owner, so a rule traces from business meaning to the stored column.	Designed	Glossary; Step 3; register (11.3)
Timeliness	Designed only. Monitoring frequency and escalation are defined. Timeliness was not measured, because Berka has no load or posting times, so the delay between an event and its recording cannot be calculated. The check that dates are not in the future (R8) is a validity check and is not evidence of timeliness.	Designed	11.5
Adaptability	A new product type is added by approving a value and adding a Product row. A change to the scope (hypothetical signatories) runs through the change process.	Designed	11.2 example; Step 3
Reporting	The scorecard reports measured quality results by rule, dimension, threshold, owner and status.	Measured	Scorecard in 11.5

Scope and limits. This project covers a small retail scope, and BCBS 239 is mainly about risk data. The note shows how the foundations support the principles. It does not claim compliance. Compliance would need an assessment of a bank's actual risk data, systems and reports, which this project does not attempt. Where the evidence is measured, it is stated as measured. Where it is design only, it is labelled as designed. The Step 4 results were measured on the Berka dataset, a public Czech bank dataset from 1999, and say nothing about any real bank's data quality. The scorecard reports those measured results, and no bank risk report was tested. Timeliness and adaptability are addressed at design level. The project supports the principles and does not claim to meet them.

Section 11: Governance operating model

11.1 Roles and responsibilities

Purpose. This section defines who is accountable for data in the model, and who performs, coordinates, approves and implements each control. Roles are described by function and are assumed, to be validated with the business.

Roles

Role	Responsibility
Data Owner	Accountable for the business definition of an entity, its acceptable quality and the decisions on remediation. **Owners do not build or load the data.** They approve what the entity means, agree its quality rules and thresholds, decide how issues are remediated, and approve changes within their domain.
Data Steward	Looks after the data day to day on the owner's behalf. Maintains definitions and the metadata register, proposes changes, monitors or coordinates quality checks, reports results to the owner and raises issues.
Data Office (central)	Maintains standards and templates, facilitates governance, keeps the glossary and register standards, and checks that the process was followed. It does not own definitions and does not approve their content.
Technical teams	Implement approved changes, build and run the automated quality checks, and enforce technical controls such as keys, types, required and unique constraints.
Data Consumer	Uses the data and reports issues to the Data Steward. Is consulted on changes that affect their use of the data.

Governance forum	Resolves disagreements between reviewers and cross-domain questions, and approves definitions or model changes that affect more than one domain. Made up of the Data Owners, with the Data Office facilitating and recording decisions.

Ownership by entity. The owner roles from the glossary act as Data Owners. A Data Steward is assigned within each owner's area. Titles are assumed and to be validated. Technical teams are not assigned by entity.

Entity	Data Owner (assumed)	Data Steward (assumed)
Party, Customer, Address	Head of Customer Onboarding / KYC	Customer data steward
Product	Head of Product Management	Product data steward
Account, Account Holder	Head of Account Services	Account data steward
Transaction	Head of Payments	Payments data steward
Channel	Head of Digital / Distribution	Channel data steward

RACI for key activities (R = responsible, A = accountable, C = consulted, I = informed). Every row has exactly one A. "A/R" means the same role is both accountable and responsible.

Activity	Data Owner	Data Steward	Data Office	Technical teams	Data Consumer	Governance forum
Define or change a term	A	R	C	I	C	I
Define or change a term used by several domains	C	R	C	I	C	A
Set a data quality rule and threshold	A	R	C	C	I	I
Run the quality checks	I	A	I	R		
Monitor results and report them to the Data Owner	I	A/R	C	C	I	
Check that the process was followed	I	C	A/R	I		
Resolve a data issue (steps in 11.7)	A	R	I	R	C	
Approve a change to the model or a constraint	A	R	C	C	I	I
Implement an approved change	I	C	I	A/R	I	
Settle a disagreement between reviewers	C	I	C	I	I	A/R

"Approve a change to the model or a constraint" is the Data Owner's decision (A), and the Data Steward proposes it (R). A change that affects several domains goes to the governance forum, as in the second row.

Constraints are enforced by the structure and changed through the model, which Technical teams implement. Data quality rules cover what the structure cannot enforce. They are owned by the Data Owner, run by Technical teams, and monitored by the Data Steward.

Limitations. Roles are assumed. A real bank's titles and structure would be confirmed with the business.

11.2 Decision rights and sign-off

Purpose. This section sets out how a change to a definition, rule or model is proposed, reviewed, approved and implemented, so that meanings stay consistent and every change is recorded.

Decision rights

Decision	Proposes	Reviews	Approves	Implements
New or changed definition	Data Steward	Data Office, affected Data Owners	Data Owner of the entity	Data Steward (glossary and register)
New or changed quality rule	Data Steward	Data Office, Technical teams	Data Owner of the entity	Technical teams
Change to the model (entity, relationship)	Data Steward	Data Office, Technical teams, affected Data Owners	Data Owner of the main entity affected	Technical teams
Definition or model change used by several domains (for example Customer)	Data Steward	All affected Data Owners, Data Office	Governance forum	Data Steward and Technical teams
New or changed value in an allowed-value list	Data Steward	Data Office	Data Owner of the entity that holds the attribute	Technical teams

New or changed constraint (required, unique, key)	Data Steward	Data Office, Technical teams	Data Owner of the entity	Technical teams

The Data Office never approves the content of a change. It checks that the process was followed.

Sign-off steps

1. **Propose.** The Data Steward records the change, the reason and the entities affected.
2. **Assess impact.** The Data Office checks the effect on the model, the glossary, the quality rules and users of the data, with Technical teams for model changes. A change to the format of an identifier or code (Customer ID, IBAN, postcode, country or currency code) always triggers a model review, because these are stored as text under decision L8.
3. **Review.** Affected Data Owners and consumers comment.
4. **Approve or reject.** The decision-maker named in the table decides and records the outcome.
5. **Validate.** The business confirms the approved wording. Until then, the owners and definitions are treated as working versions.
6. **Record and implement.** Technical teams implement the approved change. The Data Steward updates the glossary, model and register with a version number, date and approver.
7. **Communicate.** Data Consumers are informed.

Worked example from this project. The Account Holder definition originally read “the role a Customer plays on a specific Account, with rights and responsibilities”. Testing it against signatories and guarantors showed it was too broad. Under this process, the Data Steward would propose narrowing it to a Customer named on the agreement, the Data Office would check the effect on the model and rules, the Data Owner (Head of Account Services) would approve, and the glossary would be updated with a new version. The relationship wording changed in the same way: “holds” became “acts as” and later “is associated with”, and “can be a” became “can have”. The Customer wording affects a concept used across domains, so under this process it would go to the governance forum.

Hypothetical example (not part of this project’s scope). Suppose the bank wants to record authorised signatories on accounts, which this project excludes (C3). The Data Steward proposes a new entity and a link to Account, with the reason. The Data Office assesses the impact: a new entity and relationship, new glossary terms and new quality rules. This is a change to the model, so the Data Owner of the main entity affected (Head of Account Services) approves it. If the Customer definition would also change, the decision goes to the governance forum, because Customer is used across domains. Technical teams implement the change, the glossary, model and register are updated with a new version, and Data Consumers are informed.

Escalation. If reviewers disagree, the decision goes to the governance forum. The forum meets when a matter is escalated. A regular schedule is not specified in this project.

Limitations. This is a designed process and has not been run with real stakeholders.

11.3 Metadata register

Purpose. This section describes the register that holds all three kinds of metadata in one place, so each term can be traced from its business meaning to where it is stored and how its quality is checked. The register is kept in Excel. The Data Steward maintains it and the Data Office checks it.

Structure. One row per glossary attribute (17 rows), with Entity before Attribute. An Entities sheet holds the 8 entities. The attribute links the kinds of metadata.

Metadata type	**Register columns**	**Source**
Business	entity, attribute, definition, owner, example, glossary rule.	Glossary (Step 2)
Technical	table, column, data type, key, required.	Logical model (Step 3)
Operational	Berka field used, rule IDs, result, test status.	Data quality rules (Step 4)
Control	status, version, date, approver.	Sign-off step 6 (11.2)

Extract. Five of the 17 rows. The full register is in the Excel file.

Attribute	**Owner**	**Table.Column**	**Type, key**	**Quality rule**	**Status**
Account Number (IBAN)	Head of Account Services	Account.Account_Number_IBAN	text, UK	Unique, not empty	Tested on Berka, using account_id as a stand-in: 0 of 4,500 failed (R1a, R1b)
Customer ID	Head of Customer Onboarding / KYC	Customer.Customer_ID	text, PK	Unique, not empty, never reused	Enforced by key. Berka client_id profiled: 5,369 distinct of 5,369 (Script 03). "Never reused" cannot be tested on one snapshot

Transaction Amount	Head of Payments	Transaction.Transaction_Amount	decimal (2 places)	Greater than zero	Tested: 14 of 1,056,320 failed (0.0013%), all zero-amount interest postings (R2, DQ1)
Debit/Credit Indicator	Head of Payments	Transaction.Debit_Credit_Indicator	text (1)	"D" or "C" only	Target mapping tested: 0 failed (R4). Source check: 16,666 rows hold VYBER in `type`, which the documentation lists only under `operation` (F1). Mapped to D with supporting evidence (DQ2).
Channel Type	Head of Digital / Distribution	Channel.Channel_Type	text	One of the agreed values (six, including internal posting)	Rule defined. 183,114 transactions had no channel in the source. Fix designed: add 'internal posting' to the Channel Type list (DQ3), awaiting owner approval (DI-06)

Rules for keeping the register.

- **Technical key columns** (Party_ID, Transaction_ID and so on) appear on the Columns (30) sheet with the business columns, so all 30 columns and every key are shown, and they inherit their table's owner.
- **Naming convention.** Glossary attribute use business names with spaces (Account Opening Date). The logical model uses underscores (Account_Opening_Date). The register holds both, and the atribute is the link.
- **Updates.** The register changes only through the sign-off steps in 11.2. A change to a definition, type, key or rule needs a new version, date and approver.
- **Status values.** Each term is "working" until the business validates it, then "approved".
- **version line:** 0.1 on 28 October 2026 (first draft) and 0.2 on 2 November 2026 (restructured).

Limitations. The register is built by one person and the owners are assumed. The Result column is filled from the Step 4 tests on Berka, and only for the rules tested there. It does not hold operational metadata about ongoing data movement, such as load logs and monitoring. The one-off Berka row-count check in Step 4 is not a register item.

11.4 Data lifecycle and classification

Purpose. This section sets out how long each entity is kept, what happens to it afterwards, and how sensitive each attribute is. It shows that governance covers the whole life of the data, not only its definition.

Lifecycle by entity": The retention periods are assumed. They are simplified from commonly cited Dutch anti-money-laundering and tax record-keeping periods (5 and 7 years) and have not been checked against any bank's own policy.

Entity	Kept active while	After that	Retention (assumed)	Disposal
Party, Address	The Party has a relationship with the bank	Archived	5 years after the relationship ends, and not before every record that refers to it has been disposed of	Deleted or anonymised
Customer	The relationship is current	Archived, record kept	5 years after the relationship ends, and not before its Account Holder records have been disposed of	Deleted or anonymised
Product	The Product is on sale or has live Accounts	Marked retired	Kept while any Account refers to it	Retained for reference
Account	The Account is open	Kept with status "closed" (A8)	7 years after closure	Deleted or anonymised
Account Holder	The Account is open	Archived with its Account	Same as the Account	Disposed of with the Account
Transaction	Recent period, for fast access	Archived	7 years after the transaction date	Deleted or anonymised

Channel	The Channel is in use	Marked retired	Kept while any Transaction refers to it	Retained for reference

Where more than one period applies to the same data, the longer one is used. Account, Account Holder and Transaction records follow the 7-year period. Customer, Party and Address records follow the 5-year period, but Account Holder records point to Customer. A Customer with accounts is therefore disposed of only after its Account Holder records, so the 7-year period decides in practice.

Three points follow from the structure:

- A row cannot be deleted while another row points to it through a foreign key. Disposal has to follow the order Transaction, Account Holder, Account, then Customer, Address and Party.
- Closed Accounts stay in the data (A8), so "closed" starts the retention clock and does not mean deletion.
- A returning Customer keeps the same record (A3) only while that record still exists. After disposal, a returning person would be a new Party and a new Customer.

Classification by attribute. The scheme is assumed: Internal, Confidential and Restricted. "Personal data" means information about an identifiable person under GDPR.

Attribute	**Personal data?**	**Classification**	**Access (assumed)**
Party Type	Yes, when linked to a person	Confidential	Role-based
Party Name	Yes	Restricted	KYC and servicing roles
Customer ID	Yes	Confidential	Role-based
Address Type	Yes	Confidential	Role-based
Street	Yes	Restricted	KYC and servicing roles
Postcode	Yes	Restricted	KYC and servicing roles
City	Yes	Confidential	Role-based
Country	Yes	Confidential	Role-based
Product Type	No	Internal	All staff
Account Number (IBAN)	Yes	Restricted	Account servicing roles
Account Opening Date	Yes, when linked to a person	Confidential	Role-based
Account Currency	No	Internal	All staff
Account Status	Yes, when linked to a person	Confidential	Role-based

Transaction Amount	Yes, when linked to an Account Holder	Restricted	Payments and servicing roles
Debit/Credit Indicator	Yes, when linked to an Account Holder	Restricted	Payments and servicing roles
Transaction Timestamp	Yes, when linked to an Account Holder	Restricted	Payments and servicing roles
Channel Type	No	Internal	All staff

Technical key columns take the classification of their table, so they are not listed. Classification is set on the business term and shown in the register as an additional column.

Governance. The Data Owner approves the classification and retention of their entities. The Data Steward records them in the register. The Data Office checks that the process was followed. The privacy and legal functions would validate both.

Limitations.

- The classification scheme, access roles and retention periods are assumed, and no real legal or privacy review has been done. The model holds no special-category data, and cross-border rules are not considered.
- Retention periods are simplified from AML and tax rules. A real bank would set them with its legal and compliance teams, and would also check that data is not kept longer than necessary under GDPR.

11.5 Data quality KPIs

Purpose. A data quality rule says what good data looks like. A KPI says how often the data meets that rule, who is responsible for it, and what happens when it does not. This section turns the Step 4 rules into measures that an owner can be held to.

How the scorecard is built. Each row comes from a rule in the Rule column of the attribute glossary, or from a finding in Step 4, so every row traces back to a glossary term and its owner. A rule is owned by the Data Owner of the attribute that carries it in the glossary. R3 is defined under Transaction Timestamp, so Head of Payments owns it. R8 covers both Transaction Timestamp and Account Opening Date, so both owners are named. The result is the share of records that pass, taken from the results table in Section 9. R5 is shown as a review population, not a pass rate. The baselines are the Berka results. Thresholds and the monthly frequency are proposals for the owner to approve, because there is no real bank here to set them.

Data quality scorecard

Rule	Dimension	Threshold	Result (Berka)	Owner (assumed)	Status
R1a Account Number unique	Uniqueness	100%	100%	Head of Account Services	Met
R1b Account Number not empty	Completeness	100%	100%	Head of Account Services	Met
R2 Transaction amount above zero	Validity	100%	99.9987% (14 failed)	Head of Payments	Not met. Cause not confirmed. All 14 are zero-amount interest postings by k_symbol (DQ1, DI-02)
R3 Transaction not before account opening	Consistency	100%	100%	Head of Payments	Met
R4 Source direction values map to D or C	Validity (target mapping)	100%	100% after mapping	Head of Payments	Met (mapping is complete; source validity is F1)
R5 Customers with no Account Holder	Completeness	Review population reviewed within 30 days	869 (16.19%) flagged, all disponents	Head of Customer Onboarding / KYC	Review population, not an error
R6 No two transactions identical except `trans_id` (potential duplicates)	Uniqueness	100%	99.9998% of rows (1 pair, 2 rows)	Head of Payments	Not met. The pair is a potential duplicate that the data cannot resolve, and it is under review (DI-04)
R7 Balance reconciles	Consistency	100% exact (a tolerance to be agreed with the owner)	98.70% of 406,941 checked	Head of Payments	Not met, all differences 0.50 or less, cause not confirmed
R8 Dates not in the future	Validity	100%	100%	Head of Payments, Head of Account Services	Met. Timeliness is not measured
R9 Account has an Account Holder	Completeness	100%	100%	Head of Account Services	Met (Berka has one owner per account)

F1 `type` holds only documented values (PRIJEM, VYDAJ)	Validity (source)	100%	98.42% (16,666 rows hold VYBER)	Head of Payments	Not met. VYBER is documented under operation, not type. Mapped to D with supporting evidence (DQ2)
F2 Transactions have a channel	Completeness	100%	82.66% measured. 100% expected once DQ3 is approved	Head of Digital / Distribution	Fix designed: add 'internal posting' to the Channel Type (DQ3), awaiting owner approval (DI-06)

All rules are proposed for monthly measurement.

Thresholds and escalation.

- Each KPI has a target. Technical teams run the checks and the Data Steward monitors the results.
- When a result falls below its target, the Data Steward reports it to the Data Owner and logs it as an issue (11.7). The Data Owner investigates, records the cause and decides the remediation. Technical teams implement the fix.
- A proposed second line, to be agreed with the owner: a result that stays below target for two consecutive periods is raised to the governance forum.
- The Data Office checks that this process is followed.
- For R2, R6, R7 and F1, the Berka baseline already shows a failure, so the first action is to record what is known about the cause and agree what to do about it:
    - R2: zero-amount interest postings.
    - R6: a potential duplicate pair that the data cannot resolve.
    - R7: differences 0.50 or less whose cause is not confirmed. The owner also agrees whether a tolerance is acceptable.
    - F1: the owner confirms the VYBER mapping with the source owner.

Review population: Customers with no Account Holder record. This is measured as a queue, not a pass rate, because these customers are not errors (A24).

- **Owner:** Head of Customer Onboarding / KYC, who is also the owner of Customer ID.
- **Review step:** each item is checked and given one of three outcomes, recorded with a reason code:

1. The Customer legitimately has no Account Holder record, for example onboarded but not yet funded, or a role that is out of scope.
2. An Account Holder link is missing and is added.
3. The Customer record is wrong and is corrected or closed.

- **KPIs:** queue size, age of the oldest open item, and the percentage of items reviewed within the agreed time (proposed: 30 days).
- **Baseline (Berka):** 869 of 5,369 customers (16.19%), all disponents, which are out of scope (C3). So the first outcome would apply to all of them.

Limitations.

- Targets, frequencies and the 30-day review time are proposals, with no real bank data behind them.
- Baselines come from one snapshot of a 1999 dataset, so none of the KPIs shows a trend.
- Quality rules that Berka could not test (IBAN format, currency code, postcode format, Account Status) have a defined KPI in the register only when the Rule column holds a rule, and have no baseline.
- "Never reused" for Customer ID cannot be measured on a single snapshot.
- Timeliness is not measured, so there is no timeliness KPI.

11.6 Compliance and controls

Purpose. Data governance exists partly to meet regulatory obligations. This section maps three requirements to the data in the model, the control that supports each one, and the assumed owner. It shows how the definitions, ownership and quality rules built in Steps 1 to 4 support compliance. It is a mapping and does not claim that the model is compliant.

Requirements in scope. Three were chosen so that each can be explained and traced to the model:

- **GDPR (AVG in the Netherlands):** the protection of personal data.
- **KYC and anti-money-laundering requirements:** identifying customers and keeping records. In the Netherlands this is covered mainly by the Wwft (Wet ter voorkoming van witwassen en financieren van terrorisme). The exact requirements are to be confirmed with the official source.
- **BCBS 239:** principles for risk data aggregation and reporting, covered in the BCBS 239 note (Section 10).

Compliance and controls mapping

Requirement	Data it applies to	Control	Owner (assumed)
GDPR: data minimisation	Party Name, Address	Names and addresses are held on Party and Address only and are not repeated on Customer or Account Holder.	Head of Customer Onboarding / KYC
GDPR: access restriction	Personal data attributes	Each attribute is classified as personal or not personal, with an access level (see the classification table in 11.4).	Owner of the entity
GDPR: storage limitation	All entities	A retention rule is set for each entity (see the lifecycle table in 11.4). Retention periods are to be confirmed with Compliance.	Owner of the entity
KYC/AML: customer identification	Party Type, Party Name, Address	Completeness rules: not empty, and the legal name is as shown on the identity document.	Head of Customer Onboarding / KYC
KYC/AML: record keeping	Customer, Account, Transaction	A returning Customer keeps the same record (A3), and a closed Account is kept with a closed status and not deleted until its retention period ends (A8).	Head of Account Services
BCBS 239: accuracy and integrity	Transaction Amount, Account Number	Validity and uniqueness rules, tested in Step 4.	Owners of the entities tested
BCBS 239: consistency across the bank	All entities	One shared glossary with a named owner for each term.	Owner of the entity

How the controls work. Three kinds of control run through the model:

- **Definition controls.** An agreed definition and edge-case rules, so that terms such as Customer and Account Holder are read the same way everywhere.
- **Quality controls.** Rules in the attribute glossary, tested in Step 4, with results reported by data quality dimension.
- **Ownership controls.** A named owner role for each entity, accountable for its meaning and quality, and the approver of changes under the sign-off process in 11.2.

Limitations.

- The mapping shows how the model supports each requirement. It does not assess compliance, which would be a matter for Compliance and Legal.
- Requirements are summarised at a high level. Retention periods, article references and the exact scope of KYC/AML obligations are to be confirmed with the official sources and the relevant function.
- Owner roles are assumed and would be validated with the business.
- Only three requirements are covered. Others, such as payment regulation and capital reporting, are out of scope.

Feeds next. The controls feed the data quality KPIs (11.5) and the RACI (11.1), where each control has an accountable owner.

11.7 Data issue management

Purpose. This section sets out how a data problem is raised, assessed, decided and closed, so that every issue has an owner, a recorded decision and a visible status. It turns the Step 4 findings into a working issue log.

What counts as an issue.

- A rule that falls below its threshold (11.5).
- A finding from profiling, such as F1 and F2.
- A problem reported by a Data Consumer.
- A conflict between a definition and the data.

Items in the R5 review population are not issues. An item becomes an issue only when the review decides that the record is wrong (outcomes 2 and 3 in 11.5).

Lifecycle.

1. **Raise and log.** Anyone can raise an issue. The Data Steward logs it with an ID, source, description, record count and date.
2. **Triage.** The Data Steward sets the severity and confirms the Data Owner.
3. **Investigate.** The Data Steward and Technical teams establish the cause, or record that it is not established. The log states what was checked and what was not.

4. **Decide.** The Data Owner chooses one option and records the reason:
    - Correct at source.
    - Correct in the mapping or load.
    - Accept, with a reason and a review date.
    - Change the rule or definition, through the sign-off process in 11.2.
    - Refer to another owner.
5. **Remediate.** Technical teams implement the decision.
6. **Verify.** The Data Steward re-runs the rule and records the result.
7. **Close and record.** The Data Owner confirms closure, and the log holds the decision, evidence and date. An accepted issue stays in the log with its review date.

Severity and decision targets (assumed).

Severity	Criteria	Example here	Decision target
High	Wrong or unexplained values in money amounts, balances or direction, with no supporting evidence, or a result that could change a reported figure	None in this project	5 working days
Medium	A large share of records is affected, or a value is inferred from evidence, or results cannot be reproduced exactly	DI-01, DI-05, DI-06	15 working days
Low	No money affected, few records, or a documented limitation	DI-02, DI-03, DI-04, DI-07	30 days or the next monthly review

Roles. Every row has exactly one A.

Activity	Data Owner	Data Steward	Data Office	Technical teams	Data Consumer	Governance forum
Raise and log an issue	I	A/R	I	C	R	

Triage and set severity	A	R	I	C	I	
Investigate the cause	A	R	I	R	C	
Decide remediation or acceptance	A/R	C	I	C	C	
Implement the fix	I	C	I	A/R	I	
Verify and close	A	R	I	C	I	
Check the log is complete and current	I	C	A/R			
Settle a disagreement or an overdue issue	C	C	C	I	I	A/R

Escalation. An issue goes to the governance forum when:

- It stays open past its decision target.
- Owners disagree about who owns it or how to resolve it.
- It stays below target for two consecutive periods (11.5).

Measures. Open issues by severity, age of the oldest open issue, the percentage decided within target, and the number of repeat issues. All are proposals.

Issue log built from the Step 4 findings. Statuses show where each item stands in this project as at 28 October 2026. They are proposals, because no real owner has made these decisions.

ID	Source	Description	Impact	Severity	Owner (assumed)	Cause	Decision or next action	Status
DI-01	F1	16,666 transactions (1.58%) hold VYBER in type, which the documentation lists only under operation	Direction is inferred. R7 depends on it	Medium	Head of Payments	Not established	Mapped to D (DQ2). Confirm with the source owner	Open, awaiting confirmation
DI-02	R2	14 transactions have an amount of 0.00. All are interest postings by k_symbol	Breaks the rule. No money affected	Low	Head of Payments	Possibly normal zero-interest lines. Not confirmed	Owner decides: accept as an exception, or correct at	Awaiting decision

							source (DQ1)	
DI-03	Observation from R2	10 of the 14 carry the operation “cash withdrawal” although k_symbol marks them as interest	Inconsistent labelling. Checked on 14 rows only	Low	Head of Payments, with Head of Digital / Distribution consulted	Not established	Test across the whole table to find the true scope	Open, scope unknown
DI-04	R6	One pair of identical transactions (3456218 and 3507085) on account 9051	No money affected. Cannot tell if it is a duplicate	Low	Head of Payments	Cannot be determined from the data	Ask the source system. Keep as a potential duplicate	Open, awaiting source evidence
DI-05	R7	5,296 of 406,941 balances (1.30%) differ from the expected balance, all by 0.50 or less	Balances cannot be reproduced exactly in the tested subset	Medium	Head of Payments	Not established. Rounding is one possibility	Investigate. Owner decides whether a tolerance is acceptable	Open, cause not confirmed
DI-06	F2	183,114 transactions (17.34%) have no operation, so no Channel	Channel missing for bank-generated interest credits	Medium	Head of Digital / Distribution	The source leaves operation empty for these rows	Add “internal posting” to the Channel Type list (DQ3) through 11.2	Open, fix designed, awaiting approval of the new value
DI-07	Step 4 limitation	Berka has one OWNER per account, so joint accounts never occur	The many-to-many link is untested on real data	Low	Head of Account Services	Characteristic of the dataset	Accept. Retest on data with joint accounts	Accepted

Baseline. Seven issues are logged. Five are open or awaiting a decision (DI-01 to DI-05), one open with a fix designed (DI-06), and one is accepted (DI-07).

Limitations. The log was built after the fact from Step 4 findings. Owners, severities, decision targets and statuses are assumed, and no real owner has made a decision. A real log would be kept in a ticketing or governance tool, with dates for every step.

11.8 Data quality and stewardship policy and controls

Purpose. This section sets out the internal rules for how data in the model is defined, owned, quality-checked and kept, and the controls that carry those rules out. It is a draft written for this project. It has not been approved by anyone.

Scope. All eight entities and 17 attributes in the model, and the people and teams named in 11.1.

Policy statements.

#	Statement	Traces to
P1	Every business term in scope has an agreed definition, a named Data Owner and a Data Steward.	Section 6, 11.1
P2	A definition, rule, constraint or allowed value changes only through the sign-off process, and the glossary, model and register are updated with a version, date and approver.	11.2, 11.3
P3	Every attribute with a rule in the glossary has a data quality rule that is tested and measured against a threshold set by its Data Owner.	Section 9, 11.5
P4	A rule that finds no failures is shown able to detect errors before its result is relied on.	Section 9
P5	Source data is not silently changed. A value outside the source documentation is reported as a finding. A mapping decision records its evidence and does not make the source value valid.	F1, DQ2, A14
P6	Every data quality issue is logged, assigned, decided and closed under the issue process.	11.7
P7	Personal data is classified, access is role-based, and a retention period and disposal order are set for each entity.	11.4
P8	Results are reported with what they establish and what they do not: failures are kept separate from review populations, and matches are called potential duplicates until confirmed.	Section 9, 11.5

Exceptions. An exception to any statement needs written approval from the Data Owner, with a reason and a review date. It is recorded in the issue log as an accepted issue.

Ownership and review. The Data Office maintains the policy, and the governance forum would approve it. Both roles are assumed. The policy is reviewed once a year (proposed). The Data Office reports any breach of the policy to the governance forum.

Controls. Frequencies are proposals. "Status" shows what was actually done in this project.

ID	Control	Type	Policy	Frequency	Owner (assumed)	Status in this project
CTL1	One glossary with a named owner and steward for each term	Preventive	P1	On every change	Data Office maintains. Data Owners approve	Designed. Built in Section 6
CTL2	Sign-off process for changes to definitions, rules, constraints and values	Preventive	P2	On every change	Data Owner approves. Data Office checks the process	Designed, with worked examples. Not run with real stakeholders
CTL3	Structural constraints: primary key, foreign key, unique, required	Preventive	P3	Enforced at storage	Technical teams	Designed in the logical model. Not implemented in a database
CTL4	Data quality rules R1 to R9 run against thresholds	Detective	P3	Monthly	Data Owner. Data Steward monitors. Technical teams run	Run once on Berka
CTL5	Detection test: synthetic rows added to labelled copies of the tables	Detective	P4	When a rule is added or changed	Data Steward, with Technical teams	Done for R1a, R1b, R3, R4, R8 and R9
CTL6	Source-value check and recorded mapping evidence	Detective	P5	At every mapping change	Data Steward	Done for VYBER (F1, DQ2)
CTL7	Review population for customers with no Account Holder record	Detective	P6, P8	Monthly. Reviewed within 30 days	Head of Customer Onboarding / KYC	Designed, with the Berka baseline of 869
CTL8	Issue log and lifecycle	Corrective	P6	Continuous	Data Steward maintains. Data Owner decides	Designed. Log built from Step 4 findings (11.7)
CTL9	Classification, role-based access, retention and disposal order	Preventive	P7	On every change. Reviewed yearly	Data Owner approves. Privacy and legal would validate	Designed. Scheme and periods are assumed
CTL10	Scorecard reporting with the caveats stated	Detective	P8	Monthly	Data Steward	Produced once on Berka

How this relates to 11.6. CTL3, CTL4 and CTL9 are also the controls behind the GDPR, KYC/AML and BCBS 239 mappings in 11.6.

Limitations. The policy is a draft. No one has approved it, and no real function has reviewed it. Frequencies, review periods and roles are assumed, and most controls are designed and not operated.

Section 12: Limitations and future work

From Steps 1 and 2

- Owner roles and definitions are assumed, not validated with real stakeholders.
- Out of scope: signatories, guarantors, next of kin, investors, mortgages, insurance, counterparty details. A guarantor or signatory would be a Party, but the role is not modelled. Future versions could add a signatory role and other product types.
- Customer is defined broadly, so a Customer with no Account Holder record is allowed. A stricter definition would suit a narrower scope.
- Customer is the established relationship of a Party. Customer status over time (prospect, active, dormant, closed) and relationship start and end dates are not modelled (A2). A more advanced model would add a customer status or relationship history entity.
- The minimum of one Account Holder per Account is a data quality rule (R9), not a structural constraint (A4).
- Currency sits on the Account only, so foreign amounts and exchange rates are lost.
- Only the originating Channel is recorded.
- Transfers are one line per Account, so linking the two sides of a transfer is not modelled.
- Only 17 key attributes are defined, not the full set a bank would hold.

From Step 3

- All columns are required (L6), which is stricter than a real bank.
- Allowed values are rules, not reference tables (L4). A real bank would use governed code lists.
- No history: no leave-and-rejoin (A18), no status changes, no closure date (A8).
- One currency per Account, no time zones, two decimal places only.
- Transaction Type is not modelled (L9).
- The model is logical only. No physical model, indexes or performance design.

From Step 4

- Berka has no times, IBAN or currency, so timestamp rules run at date level and the IBAN and currency rules could not run. Party Name, Address, Product Type and Account Status were not tested.
- Timeliness, meaning the delay between an event and its recording, was not measured.
- Every Berka account has exactly one OWNER, so joint accounts and the many-to-many Account Holder link are never exercised.

- R7 covers 38.5% of transactions, and it relies on the VYBER mapping.
- R4 shows that the mapping is complete, not that the source values are valid. The VYBER mapping (DQ2) is an inference supported by evidence, to be confirmed with the source owner.
- The R6 pair is a potential duplicate that the data cannot resolve. The cause of the R7 differences is not confirmed.
- The dataset is from 1999 and says nothing about any real bank.
- Ten of the 17 glossary attributes could not be tested, because Berka has no matching field: Party Type, Party Name, Address Type, Street, Postcode, City, Country, Product Type, Account Currency and Account Status. Customer ID was profiled but not tested, and Account Number was tested using account_id as a stand-in.

From the governance model (Section 11)

- Roles, thresholds, review times, retention periods and the classification scheme are assumed and have not been reviewed by Compliance, Legal or Privacy.
- The decision process has not been run with real stakeholders.
- The register is built by one person.
- The Wwft and other regulatory details are to be confirmed with the official sources.

Future work. Add reference tables, optional columns, history (start and end dates), customer lifecycle status, a Transaction Type, and a physical model. Test the rules on data with joint accounts and with times. Measure timeliness. Validate roles, owners and retention periods with the business.

From issue management and policy (11.7 and 11.8)

- The issue log was built after the fact from Step 4 findings. Severities, decision targets and statuses are assumed, and no real owner has decided any issue.
- The policy is a draft with no approver. Most controls are designed and have not been operated.
- A real bank would keep issues in a ticketing or governance tool, with dates for every step.

Section 13: Conclusion

This project followed business concepts from an agreed definition, to a model, to a testable data quality rule, with a named owner accountable for the meaning.

One concept shows the whole path. Transaction Amount is defined in the glossary as the value of money moved, always positive and in the Account's currency, and the Head of Payments is its assumed owner. The conceptual model places it on Transaction. The logical model stores it as a required decimal with two places. The rule "greater than zero" tests it in Step 4. On the Berka data, 14 of 1,056,320 amounts failed, all of them zero-amount interest postings. Under the governance model, the owner decides whether these are corrected at source or accepted (DQ1, Section 11.5).

The work showed three things:

- **Definitions shape the model.** Testing definitions against edge cases changed the model. Account Holder was narrowed to exclude signatories and guarantors. Customer was redefined as a relationship attached to a Party. Two relationship verbs were reworded.
- **A passing rule does not prove the data is valid.** The mapping rule R4 passed, while F1 showed that 16,666 source values were undocumented. Rules that found no failures were shown able to detect errors on synthetic rows. A review population (R5) was kept separate from failures.
- **Quality results need owners.** The scorecard, decision rights and sign-off process turn a failure count into a decision that a named role is accountable for.

This is a learning and demonstration project on a public dataset from 1999. It supports the BCBS 239 principles but does not claim compliance, and it says nothing about the data quality of any real bank. The limitations and next steps are in Section 12.